Experiment 6

Aim: Apply K-Means and Hierarchical Clustering on sample datasets.

Problem statement: Businesses need to understand customer behavior to improve marketing strategies and increase revenue. Customer segmentation helps in grouping similar customers based on their purchasing patterns.
This experiment aims to:

- Apply K-Means Clustering for customer segmentation
- Apply Hierarchical Clustering for comparison
- Visualize customer groups based on spending habits and income

Dataset Link:

Mall Customers Dataset
https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python

Dataset Description

The dataset contains customer information collected from a shopping mall.

- **Features include:**
  - Customer ID
  - Gender
  - Age
  - Annual Income (k$)
  - Spending Score (1–100)
- **Selected Features for Clustering:**
  - Annual Income
  - Spending Score
- **Dataset Characteristics:**
  - Unsupervised dataset (no target variable)
  - Used for pattern discovery and segmentation
  - Suitable for clustering algorithms

Mathematical Formulation of the Algorithm

K-Means Objective Function

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2$$

- Minimizes distance between data points and their cluster centroids

Hierarchical Clustering (Ward's Method)

$$D(A,B) = \frac{|A||B|}{|A|+|B|}||\mu_A - \mu_B||^2$$

- Merges clusters based on minimum variance

Algorithm Limitations

K-Means

- Requires predefined number of clusters (K)
- Sensitive to initial centroid selection
- Affected by outliers

Hierarchical Clustering

- Computationally expensive for large datasets
- Difficult to decide optimal number of clusters
- Cannot be easily reversed once merged

Methodology:

1. Data Collection
The Mall Customers dataset was obtained from Kaggle. It contains customer details such as Customer ID, Gender, Age, Annual Income, and Spending Score.
Since this is an unsupervised learning problem, there is no target variable, and the goal is to identify hidden patterns in customer behavior.
2. Data Preprocessing
- The dataset was loaded and inspected for structure and missing values.
- Only two relevant features were selected for clustering:
  - Annual Income (k$)
  - Spending Score (1–100)
- These features were chosen because they best represent customer purchasing behavior.
- Feature scaling was applied using StandardScaler to normalize the data and ensure equal contribution of variables during distance calculation.

3. K-Means Clustering Methodology
- K-Means is a partition-based clustering algorithm.

- The optimal number of clusters was determined using the Elbow Method.
- Within-Cluster Sum of Squares (WCSS) was calculated for different values of K (1–10).
- The point where the reduction in WCSS slows significantly was selected as the optimal K.
- The dataset was then clustered into K groups, and each data point was assigned to its nearest centroid.

4. Hierarchical Clustering Methodology

- Hierarchical clustering is a tree-based clustering method.
- Agglomerative (bottom-up) approach was used.
- Initially, each data point is considered as an individual cluster.
- Clusters are merged step-by-step based on minimum distance.
- A dendrogram was plotted using Ward's linkage method to visualize cluster formation.
- The optimal number of clusters was selected by analyzing the dendrogram.

5. Visualization

- K-Means clusters were visualized using scatter plots with different colors representing different clusters.
- Centroids were plotted to show cluster centers.
- Hierarchical clustering results were visualized using both dendrogram and scatter plots.

Performance Analysis

- Compared clustering results visually

Observations

- K-Means produced well-defined clusters
- Hierarchical clustering provided detailed cluster relationships
- Both methods identified similar customer segments
- Visualizations used:
    - Elbow Curve (K vs WCSS)
    - Cluster Scatter Plots
    - Dendrogram

Hyperparameter Tuning

K-Means

- Tuned parameter: **Number of clusters (K)**
- Selected using Elbow Method

Hierarchical Clustering

- Tuned parameter: **Linkage method (Ward)**

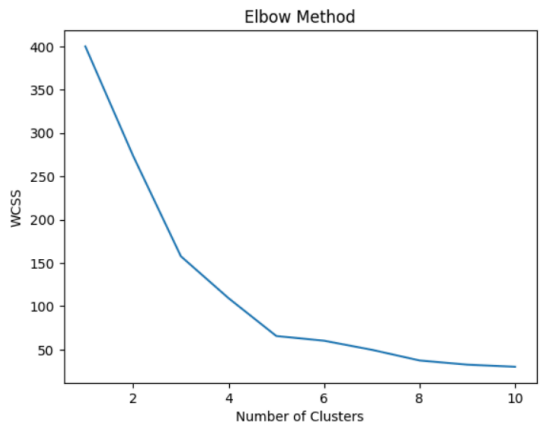
- Cluster count determined using dendrogram

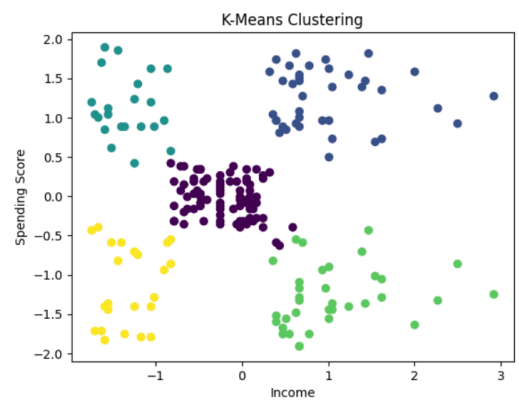
Impact

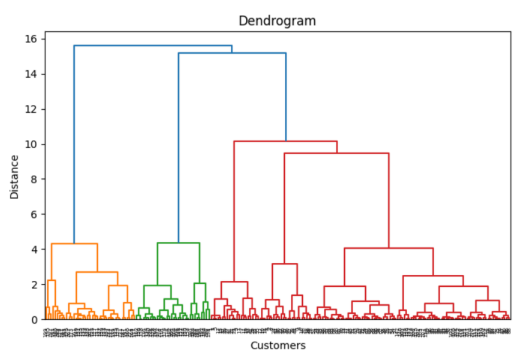
- Improved clustering accuracy
- Better segmentation of customer groups
- Clear visualization of patterns

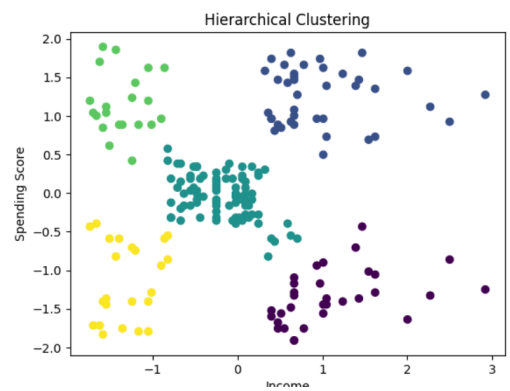
Result:

K Means









SDG 8: Improves business growth through customer segmentation
SDG 9: Applies clustering techniques for data-driven insights
SDG 12: Promotes efficient resource usage based on customer behavior